

Visitor Management Backend - Technical Documentation

Spring Boot 4.1.0 · Java 21 · PostgreSQL 17 · Hibernate 7.4

1. Architecture Overview

+-----+ REST Controllers +-----+	
TimeSlotController · VisitorController	
RegistrationController · BriefingController	
EventController · FeedbackController	
+-----+	
Services	
SchedulingServiceImpl · RegistrationServiceImpl	
EventServiceImpl · FeedbackServiceImpl	
SchedulingRules (constants)	
+-----+	
Repositories (JPA)	
TimeSlotRepository · VisitorRepository	
RegistrationRepository · BriefingRepository	
EventRepository · FeedbackRepository	
SurveySendRepository	
+-----+	
JPA Entities	
TimeSlot · Visitor · Registration · Briefing · Event	
Feedback · SurveySend	
(extend BaseEntity: id + createdAt + updatedAt)	
+-----+	
core/ - shared infrastructure	
BaseEntity · GlobalExceptionHandler · ApiError	
ResourceNotFoundException · ConflictException	
BusinessException	
+-----+	

Packages:

- core/ - shared infra only (BaseEntity, exceptions, handler)
- modules/visitormanagement/ - Scheduling + Registration sub-modules
- Enums live in the module package (not core/)

2. Database Schema

Entity-Relationship Diagram

+-----+ time_slot +-----+		+-----+ visitor +-----+	
id BIGINT PK		id BIGINT PK	
slot_date DATE NOT NULL		full_name VARCHAR(150)	NOT NULL
start_time TIME NOT NULL		group_size INT	NOT NULL, DEFAULT 1
end_time TIME NOT NULL		email VARCHAR(200)	UNIQUE
max_capacity INT NOT NULL		phone VARCHAR(30)	
status VARCHAR(20) NOT NULL		language VARCHAR(40)	
guide_id BIGINT		visitor_type VARCHAR(20)	NOT NULL
created_at INSTANT NOT NULL		special_needs VARCHAR(500)	
updated_at INSTANT NOT NULL		created_at INSTANT	NOT NULL
+-----+		+-----+	
1		1	
*		1	*

```

+-----+-----+-----+
|               registration               |
+-----+-----+-----+
| id            BIGINT| PK                |
| visitor_id    BIGINT| FK -> visitor (NOT NULL)             |
| time_slot_id  BIGINT| FK -> time_slot (nullable, NULL for    |
|               | event-only registrations)      |
| event_id      BIGINT|                |
| visit_purpose   VARCHAR(20)| NOT NULL          |
| is_prospect   BOOLEAN| NOT NULL, DEFAULT false |
| status        VARCHAR(20)| NOT NULL, DEFAULT 'PENDING' |
| created_at    INSTANT| NOT NULL          |
| updated_at    INSTANT| NOT NULL          |
| UNIQUE (visitor_id, time_slot_id)      |
+-----+-----+-----+
| 1 |
| 1 |
| 1 |
+-----+-----+
|   briefing   |
+-----+-----+
| id            BIGINT| PK                |
| registration_id BIGINT| FK -> registration |
|               | (NOT NULL, UNIQUE) |
| delivered_at  INSTANT|                |
| staff_member  VARCHAR(150)|                |
| signature     VARCHAR(500)|                |
| status        VARCHAR(20)| NOT NULL, 'PENDING' |
| created_at    INSTANT| NOT NULL          |
| updated_at    INSTANT| NOT NULL          |
+-----+-----+

```

registration.event_id -> event.id (no FK constraint, soft link)

```

+-----+-----+-----+
|               event               |
+-----+-----+-----+
| id            BIGINT| PK                |
| title         VARCHAR(150)| NOT NULL          |
| event_type    VARCHAR(30)| NOT NULL          |
| start_date_time TIMESTAMP| NOT NULL          |
| end_date_time  TIMESTAMP| NOT NULL          |
| max_capacity   INT| NOT NULL, DEFAULT 10|
| status        VARCHAR(20)| NOT NULL, 'DRAFT' |
| description    VARCHAR(1000)|                |
| location      VARCHAR(150)|                |
| created_at    INSTANT| NOT NULL          |
| updated_at    INSTANT| NOT NULL          |
+-----+-----+

+-----+-----+-----+
|               feedback               |
+-----+-----+-----+
| id            BIGINT| PK                |
| visitor_id    BIGINT| FK -> visitor (NOT NULL)|
| survey_send_id BIGINT| FK -> survey_send (nullable)|
| origin        VARCHAR(20)| NOT NULL          |
| rating        INT| NOT NULL (1..5) |
| briefing_clear VARCHAR(1000)|                |
| educational_value VARCHAR(1000)|                |
| recommend     VARCHAR(1000)|                |
| comment       VARCHAR(1000)|                |
| routed_to     VARCHAR(60)|                |

```

```

| submitted_at INSTANT| NOT NULL      |
| created_at   INSTANT| NOT NULL      |
| updated_at   INSTANT| NOT NULL      |
+-----+

+-----+
|          survey_send          |
+-----+
| id          BIGINT| PK          |
| visitor_id   BIGINT| FK -> visitor (NOT NULL)|
| channel     VARCHAR(20)| NOT NULL    |
| message_template VARCHAR(500)|          |
| sent_at     INSTANT| NOT NULL    |
| status      VARCHAR(20)| NOT NULL, 'SENT'|
| created_at   INSTANT| NOT NULL    |
| updated_at   INSTANT| NOT NULL    |
+-----+

```

Table Summary

Table	Rows (design)	Purpose
`time_slot`	~500/year	Farm tour calendar slots (2/day max)
`visitor`	~2,000	Visitor database (individuals/groups)
`registration`	~4,000	Links visitor -> time slot **or** event
`briefing`	~3,000	Safety briefing per confirmed registration
`event`	~50/year	Open days, buyer visits, school/community days
`feedback`	~3,000	Satisfaction responses (tablet + survey links)
`survey_send`	~8,000	Post-visit surveys sent by email/SMS/WhatsApp

Enums

Enum	Values	Table Column
`TimeSlotStatus`	`AVAILABLE`, `RESERVED`, `FULL`, `CANCELLED`, `COMPLETED`	`time_slot.status`
`RegistrationStatus`	`PENDING`, `CONFIRMED`, `REJECTED`, `CHECKED_IN`, `CANCELLED`	`registration.status`
`BriefingStatus`	`PENDING`, `DONE`	`briefing.status`
`VisitorType`	`INDIVIDUAL`, `GROUP`, `SCHOOL`, `PARTNER`	`visitor.visitor_type`
`VisitPurpose`	`TOURISM`, `PURCHASE`, `PARTNERSHIP`, `INVESTMENT`, `EDUCATION`, `OTHER`	`registration.visit_purpose`
`EventType`	`OPEN_DAY`, `PARTNER_BUYER`, `SCHOOL`, `COMMUNITY`, `OTHER`	`event.event_type`
`EventStatus`	`DRAFT`, `PUBLISHED`, `CANCELLED`, `COMPLETED`	`event.status`
`FeedbackChannel`	`ON_SITE`, `EMAIL`, `SMS`, `WHATSAPP`	`feedback.origin`, `survey_send.channel`
`SurveyStatus`	`SENT`, `RECEIVED`	`survey_send.status`

3. REST API Endpoints

Base URL: <http://localhost:8080/api/v1>

3.1 Time Slots - /api/v1/time-slots

Method	Path	Description	Status Codes
`GET`	`/time-slots`	List all slots (optional `?date=YYYY-MM-DD`)	200
`GET`	`/time-slots/{id}`	Get slot by ID	200, 404
`POST`	`/time-slots`	Create a new slot	201, 400, 409, 422
`PUT`	`/time-slots/{id}`	Update slot details	200, 404, 422
`DELETE`	`/time-slots/{id}`	Cancel a slot (soft-delete)	204, 404, 422
`POST`	`/time-slots/{id}/assign-guide`	Assign a guide (`?guideId=N`)	200, 404, 422
`GET`	`/time-slots/availability?date=...`	Availability view for a date	200

```

{
  "date": "2026-09-08",
  "startTime": "09:00",

```

```

"endTime": "11:00",
"maxCapacity": 10,
"guideId": 5
}

```

Create/Update Request Body:

```

{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "booked": 3,
  "status": "RESERVED",
  "guideId": 5,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}

```

TimeSlotResponse:

```

{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "remaining": 7,
  "status": "RESERVED",
  "guideId": 5
}

```

AvailabilityResponse:

3.2 Visitors - /api/v1/visitors

Method	Path	Description	Status Codes
GET	/visitors	List all visitors	200
GET	/visitors/{id}	Get visitor by ID	200, 404
POST	/visitors	Create visitor	201, 400, 409
PUT	/visitors/{id}	Update visitor	200, 404, 409

```

{
  "fullName": "Alice Dupont",
  "groupSize": 2,
  "email": "alice@example.com",
  "phone": "+241 06 12 34 56",
  "language": "French",
  "type": "INDIVIDUAL",
  "specialNeeds": "Wheelchair access required"
}

```

VisitorRequest:

3.3 Registrations - /api/v1/registrations

Method	Path	Description	Status Codes
GET	/registrations	List (filter by `?timeSlotId=`, `?date=`, or `?prospect=true`)	200
GET	/registrations/{id}	Get registration by ID	200, 404
POST	/registrations	Create registration (visitor->slot, requires `visitPurpose`)	201, 400, 409, 422
PATCH	/registrations/{id}/approve	Approve -> triggers briefing creation	200, 404, 422
PATCH	/registrations/{id}/reject	Reject a pending registration	200, 404, 422
PATCH	/registrations/{id}/check-in	Check in a confirmed visitor	200, 404, 422
PATCH	/registrations/{id}/cancel	Cancel a registration	200, 404, 422

```
{
  "visitorId": 1,
  "timeSlotId": 10,
  "visitPurpose": "PURCHASE",
  "eventId": null
}
```

RegistrationRequest:

```
{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "groupSize": 2,
  "timeSlotId": 10,
  "slotDate": "2026-09-08",
  "slotStart": "09:00",
  "eventId": null,
  "visitPurpose": "PURCHASE",
  "prospect": true,
  "status": "PENDING",
  "briefingId": null,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

RegistrationResponse (new fields):

3.4 Briefings - /api/v1

Method	Path	Description	Status Codes
GET	/registrations/{registrationId}/briefing	Get briefing for registration	200, 404
PATCH	/registrations/{registrationId}/briefing/deliver	Mark briefing delivered	200, 422

```
{
  "staffMember": "Guard Camille",
  "signature": "digital-signature-or-name"
}
```

BriefingDeliverRequest:

3.5 Events - /api/v1/events

Method	Path	Description	Status Codes
GET	/events	List events (`?type=`, `?date=YYYY-MM-DD`)	200
GET	/events/{id}	Get event by ID	200, 404
POST	/events	Create event (default status `DRAFT`)	201, 400, 422
PUT	/events/{id}	Update event (not allowed when CANCELLED/COMPLETED)	200, 404, 422
DELETE	/events/{id}	Cancel event (soft-delete)	204, 404, 422
POST	/events/{id}/publish	Publish a DRAFT event	200, 404, 422
GET	/events/{id}/registrations	List registrations for the event	200, 404
POST	/events/{id}/register	Register a visitor for the event	201, 400, 404, 409, 422

```
{
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall"
}
```

EventRequest:

```
{
  "id": 1,
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "booked": 3,
  "status": "PUBLISHED",
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall",
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

EventResponse:

```
{
  "visitorId": 1,
  "visitPurpose": "PURCHASE"
}
```

EventRegistrationRequest:

The registration created for an event has timeSlotId: null and carries the visitor's visitPurpose. Commercial purposes (PURCHASE, PARTNERSHIP, INVESTMENT) automatically mark the visitor as a prospect, exactly like time-slot registrations.

3.6 Feedback & Surveys - /api/v1

Method	Path	Description	Status Codes
GET	/feedback	List responses (`?visitorId=`, `?from=`, `?to=`)	200
GET	/feedback/summary	Compiled stats (`?from=`, `?to=`)	200
POST	/feedback	Submit response (on-site or linked to a sent survey)	201, 400, 404, 422
PATCH	/feedback/{id}/route	Route a response to a module owner	200, 404, 400
GET	/surveys	List sent surveys (`?visitorId=`, `?status=`)	200
POST	/surveys	Send a post-visit survey (email/SMS/WhatsApp)	201, 400, 404, 422

```
{
  "visitorId": 1,
  "surveyId": 12,
  "rating": 4,
  "briefingClear": "Yes, very clear",
  "educationalValue": "The solar tracking was interesting",
  "recommend": "Definitely",
  "comment": "Would love a German-language brochure"
}
```

FeedbackRequest: visitorId (required), surveyId (optional - links the response to a sent survey and marks it RECEIVED), rating (1-5, required), briefingClear, educationalValue, recommend, comment.

```
{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "surveyId": 12,
  "origin": "EMAIL",
  "rating": 4,
  "briefingClear": "Yes, very clear",
  "educationalValue": "The solar tracking was interesting",
  "recommend": "Definitely",
  "comment": "Would love a German-language brochure",
  "routedTo": "Sales (Mariata)",
  "submittedAt": "2026-08-26T13:00:00Z",
  "createdAt": "2026-08-26T13:00:00Z",
  "updatedAt": "2026-08-26T13:00:00Z"
}
```

```
}
```

FeedbackResponse:

```
{
  "averageRating": 4.2,
  "total": 18,
  "recommendPct": 92.0,
  "distribution": { "1": 0, "2": 0, "3": 3, "4": 6, "5": 9 }
}
```

FeedbackSummaryResponse (compiled weekly report):

```
{
  "visitorId": 3,
  "channel": "EMAIL",
  "messageTemplate": "Thank you for visiting Sustainable Farm! Share your feedback: [link]"
}
```

SurveyRequest:

SurveyResponse: id, visitorId, visitorName, channel, messageTemplate, sentAt, status (SENT/RECEIVED), rating (once the visitor responded), createdAt, updatedAt.

On-site tablet submissions use POST /feedback without surveyId -> origin: "ON_SITE". A sent survey becomes RECEIVED automatically when a response is linked to it.

4. Business Rules

Scheduling Rules (SchedulingRules.java)

Rule	Value	Source
Max visitors per slot	**10**	Confirmed with module owners (Abdoul)
Max slots per day	**2**	Mockup: morning + afternoon
Closed day	**Sunday**	All module owners
Morning slot	09:00-11:00	Default
Afternoon slot	14:00-16:00	Default

Registration Rules

Rule	Enforcement
Cannot register on a **CANCELLED** slot	`BusinessRuleException`
No duplicate registration (visitor + slot)	`ConflictException` (409)
Group size must fit remaining capacity	`BusinessRuleException`
Only **PENDING** -> approve/reject	`BusinessRuleException`
Only **CONFIRMED** -> check-in	`BusinessRuleException`
Cannot cancel after **CHECKED_IN**	`BusinessRuleException`
CONFIRMED -> auto-creates `Briefing` (PENDING)	Service layer
Briefing delivery requires PENDING status	`BusinessRuleException`

Event Rules

Rule	Enforcement
`endTime` must be after `startTime`	`BusinessRuleException`
Only **DRAFT** events can be published	`BusinessRuleException`
Cannot publish an event that has already ended	`BusinessRuleException`
Cannot update **CANCELLED** or **COMPLETED** events	`BusinessRuleException`
Cannot cancel a **COMPLETED** event	`BusinessRuleException`
Registration only on **PUBLISHED** events	`BusinessRuleException`
No duplicate registration (visitor + event)	`ConflictException` (409)
`booked + groupSize > maxCapacity` -> reject	`BusinessRuleException` (422)
Booked count excludes `REJECTED` / `CANCELLED` registrations	Repository query

Feedback & Survey Rules

Rule	Enforcement
Rating must be between 1 and 5	Bean validation (400)
Response linked to a survey must match the survey's visitor	`BusinessRuleException` (422)
A survey can only be answered once (RECEIVED is final)	`BusinessRuleException` (422)
Survey channel cannot be `ON_SITE` (tablet responses are direct)	`BusinessRuleException` (422)
Response without a survey -> recorded as `ON_SITE`	Service layer
Linking a response marks the survey `RECEIVED`	Service layer
Routing targets are free-text tags (no coupling to other modules)	Data model

Visitor Rules

Rule	Enforcement
Duplicate email -> `ConflictException` (409)	On create and update
`@NotBlank` on fullName	Bean validation (400)
`@Email` on email	Bean validation (400)

Lead Generation (Prospect Detection)

The visitor management app doubles as a marketing tool for the Sales & Marketing team. When a visitor registers with a commercial purpose, the system automatically flags them as a prospect.

How it works:

1. During registration, the visitor must select a visitPurpose (mandatory field)
2. If the purpose is PURCHASE, PARTNERSHIP, or INVESTMENT, the system sets isProspect = true automatically
3. Sales & Marketing can query all prospects via GET /api/v1/registrations?prospect=true

VisitPurpose values:

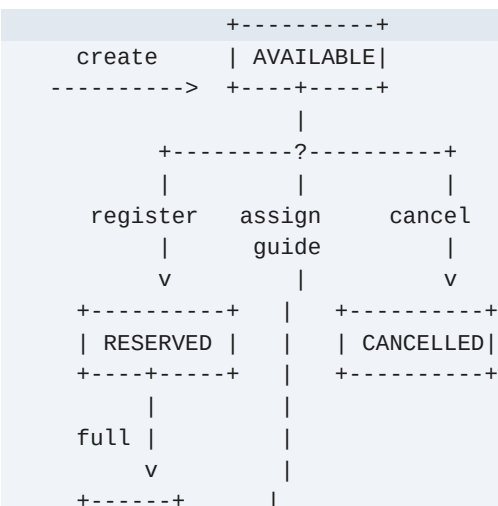
Purpose	Prospect?	Description
`TOURISM`	No	Leisure visit, tourism
`PURCHASE`	**Yes**	Interested in buying products
`PARTNERSHIP`	**Yes**	Business partnership opportunity
`INVESTMENT`	**Yes**	Investment opportunity
`EDUCATION`	No	Training, educational visit
`OTHER`	No	Other reasons

GET /api/v1/registrations?prospect=true

Prospect query example: Returns all registrations where isProspect = true, including visitor name, email, phone, and visit purpose.

5. Status Flow Diagrams

TimeSlot Status Flow




```

| FULL |<-----+ (booked >= capacity)
+-----+
|
end_time passes
v
+-----+
| COMPLETED |
+-----+

```

Registration Status Flow

```

register      +-----+
              | PENDING |
----->      +-----+
              |
              +-----?-----+
              |       |       |
            approve reject  cancel
              |       |       |
              v       v       v
+-----+ +-----+ +-----+
|CONFIRMED| | REJECTED | | CANCELLED |
+-----+ +-----+ +-----+
|
check-in
|
v
+-----+
|CHECKED_IN|
+-----+
|
cancel blocked
|
(final state)

```

Briefing Status Flow

```

auto      +-----+
          | PENDING | <- created when registration CONFIRMED
----->   +-----+
          |
          deliver briefing
          |
          v
+-----+
|  DONE  | <- staffMember + signature + deliveredAt
+-----+

```

Event Status Flow

```

create      +-----+
            | DRAFT  |
----->     +-----+
            |
            publish (not ended)
            |
            v
+-----+      cancel from any non-completed
| PUBLISHED |<-----+
+-----+      |
ended        |      +-----+
            |      | CANCELLED |
            |      +-----+
            v
+-----+

```

```
| COMPLETED |
+-----+
```

Survey Status Flow

```
+-----+
POST | SENT | <- survey link dispatched by email/SMS/WhatsApp
-----> +-----+
      |
      | response linked
      |
      | v
+-----+
| RECEIVED | <- linked Feedback (rating visible)
+-----+
```

6. Error Handling

All errors return a consistent JSON envelope via GlobalExceptionHandler:

```
{
  "status": 404,
  "error": "Not Found",
  "message": "Time slot 999 not found",
  "path": "/api/v1/time-slots/999"
}
```

Exception	HTTP Status	When
`ResourceNotFoundException`	404	Entity not found
`ConflictException`	409	Duplicate email, duplicate registration, overlapping slot
`BusinessRuleException`	422	Invalid state transition, capacity exceeded, Sunday creation
`MethodArgumentNotValidException`	400	Bean validation failures (`@NotBlank`, `@Email`, etc.)

7. Test Coverage Summary

Layer	Annotation	Tests	Framework
Repository	`@DataJpaTest` + H2	27	Spring Data + AssertJ
Service	`@ExtendWith(MockitoExtension.class)`	62	Mockito + AssertJ
Controller	`@WebMvcTest` + MockMvc	57	MockMvc + Mockito
Context	`@SpringBootTest`	1	Spring Boot
Total		**147**	

Test Files

```
src/test/java/com/infineonbit/sustainablefarm/
+-- SustainableFarmApplicationTests.java
+-- modules/visitorsmanagement/
    +-- repository/
        | +-- TimeSlotRepositoryTest.java      (5 tests)
        | +-- VisitorRepositoryTest.java       (4 tests)
        | +-- RegistrationRepositoryTest.java  (5 tests)
        | +-- BriefingRepositoryTest.java      (2 tests)
        | +-- EventRepositoryTest.java         (4 tests)
        | +-- FeedbackRepositoryTest.java      (7 tests)
    +-- service/
        | +-- SchedulingServiceImplTest.java   (13 tests)
        | +-- RegistrationServiceImplTest.java (21 tests)
        | +-- EventServiceImplTest.java        (16 tests)
        | +-- FeedbackServiceImplTest.java     (12 tests)
    +-- controller/
        +-- TimeSlotControllerTest.java        (9 tests)
        +-- VisitorControllerTest.java         (7 tests)
        +-- RegistrationControllerTest.java     (12 tests)
        +-- BriefingControllerTest.java        (5 tests)
```

```
+- - EventControllerTest.java      (13 tests)
+- - FeedbackControllerTest.java   (10 tests)
```

8. Running the Backend

```
# Start PostgreSQL + Backend
docker compose up -d backend

# Backend available at
http://localhost:8080

# Swagger UI
http://localhost:8080/swagger-ui/index.html

# Actuator health
http://localhost:8080/actuator/health

# Run tests (Docker, no local JDK required)
docker run --rm -v "$PWD":/src -w /src \
  -v "$HOME/.m2":/root/.m2 \
  maven:3.9-eclipse-temurin-21 ./mvnw test

# Compile only
docker run --rm -v "$PWD":/src -w /src \
  -v "$HOME/.m2":/root/.m2 \
  maven:3.9-eclipse-temurin-21 ./mvnw compile
```

9. Technology Stack

Component	Version	Notes
Java	21	LTS
Spring Boot	4.1.0	Modular monolith
Spring Data JPA	4.1.0	Derived queries + `@Query`
Hibernate ORM	7.4.1	`ddl-auto=update` (no Flyway)
PostgreSQL	17-alpine	Production database
H2	2.x	Test-only in-memory DB
SpringDoc OpenAPI	2.8.5	Swagger UI auto-generated
Lombok	1.18.46	Boilerplate reduction
Jackson	3.1.4	JSON serialization
Mockito	5.23.0	Service + controller tests
AssertJ	3.27.7	Fluent assertions

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